A Factorized Oscillatory-Envelope Derivative Calculus for Local Turning Models in Machine Computation

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Abstract

Local turning analysis for machine computation can benefit from derivative structure rather than only path approximation. This paper develops a factorized oscillatory-envelope derivative calculus for sparse local nodes on a short forward computational window. A local node has the form

$$Z(\tau) = b\tau^m e^{\alpha(\tau)} e^{\mathrm{i}(\omega\tau+\phi)}, \qquad 0 < \tau \le T,$$

where the polynomial scale, envelope, and oscillatory phase represent distinct local shape components. All quotient identities are stated pointwise on admissible evaluation sets, so singular behavior at the unshifted origin remains explicit. The central identity is the derivative factorization

$$Z^{(n)}(\tau) = Z(\tau)P_n(\tau),$$

where the derivative symbol $P_n = Z^{(n)}/Z$ is generated by the normalized derivative recursion $P_{n+1} = P_n' + qP_n$ with

$$q(\tau) = m/\tau + \alpha'(\tau) + \mathrm{i}\omega.$$

Equivalently, P_n is the complete exponential Bell polynomial in $q, q', \ldots, q^{(n-1)}$. The quotient form is emphasized because expanded complex formulas may contain hidden sign alternation and cancellation after real projection. The computational object is therefore the multiplicative pair (Z, P_n), or equivalently the log-polar node together with normalized derivative ratios. In the constant-envelope-slope case $\alpha'(\tau) = r$, the derivative hierarchy reduces to a binomial falling-factorial formula

$$Z^{(n)}(\tau) = Z(\tau)\sum_{k=0}^{n}\binom{n}{k}\frac{(m)_k}{\tau^k}(r+\mathrm{i}\omega)^{n-k}.$$

For the default computational package, m is kept as a small nonnegative integer. Non-integer or decimal exponents are possible through generalized falling factorials, but they are used with either an origin-excluding window $[\tau_{\min}, T]$ or a shifted coordinate $\tau + \delta$ to control singular derivative ratios. The paper also records an algebraically equivalent imaginary-axis coordinate package obtained from $\zeta = \mathrm{i}\tau$, interpreted as a finite-order contour-coordinate derivative rather than as an unrestricted holomorphic derivative. The resulting first, second, and third derivative ratios are used to define local direction, curvature, and curvature-momentum blocks. A reference implementation checks the analytic quotient-symbol recursion, coordinate equivalences, finite-difference validation behavior, derivative-aware fitting protocol, and controlled high-noise synthetic stress tests. These computational tests support the numerical consistency of the derivative package, but they are not presented as real-market predictive validation.

Keywords: factorized derivative calculus; oscillatory-envelope nodes; logarithmic derivatives; Bell polynomials; falling factorials; local turning models; derivative-aware fitting; synthetic stress tests; machine computation; numerical stability.

JEL classification: C02; C63; C65.

1 Introduction

Local turning analysis appears in computational models whenever a short path segment may change direction before a global structure has been resolved. A global approximation model may be too large or too global for this task, while an ordinary local polynomial fit may capture position but miss compact oscillatory geometry. This paper develops Factorized Derivative Calculus for Local Turning (FDC-LT) as a theoretical framework for machine computation. Its objective is to build a compact derivative package that can be evaluated, normalized, and reused by an algorithm.

The time coordinate is shifted to a current reference point t_0 by

$$\tau = t - t_0, \qquad 0 \le \tau \le T,$$

where T is a short visible computational horizon. The object is not a direct financial-market engine and not a global reconstruction of an entire path. The object is local sensing on $[0, T]$, especially the extraction of first-, second-, and third-order derivative information. The interpretation is

$$\begin{aligned} x'(\tau) &\quad \text{local direction,} \\ x''(\tau) &\quad \text{local curvature,} \\ x'''(\tau) &\quad \text{curvature momentum.} \end{aligned}$$

The third quantity is included because a developing bend is not only a high-curvature event; it may also be a region where curvature is changing quickly.

The basic local oscillatory-envelope node is

$$Z(\tau) = b\tau^m e^{\alpha(\tau)} e^{\mathrm{i}(\omega\tau+\phi)}. \tag{1}$$

The coefficient b is the node amplitude factor, m is the local scale exponent, α is the envelope, ω is the local frequency, and ϕ is the phase. The factor τ^m controls local polynomial scaling, $e^{\alpha(\tau)}$ controls mild envelope deformation, and $e^{\mathrm{i}(\omega\tau+\phi)}$ carries local phase geometry. A real-valued local signal can be obtained from real and imaginary channels or from real linear combinations of complex nodes.

A computational issue arises immediately. The symbolic formulas may appear to contain only additions, but complex exponentials and real-channel projections necessarily introduce sign alternation. For example, powers of $\mathrm{i}\omega$ generate negative real terms, and sums of out-of-phase nodes may cancel. Thus an implementation that expands all formulas into additive real coordinates too early may be less stable than the symbolic formula suggests.

The methodological response is not to eliminate addition completely. Derivative algebra contains addition through the product rule, the Leibniz rule, and Bell-polynomial structure. The response is instead to use a multiplicative quotient representation. Every derivative of a node is written as the node itself multiplied by a derivative symbol,

$$Z^{(n)}(\tau) = Z(\tau)P_n(\tau), \qquad P_n(\tau) = \frac{Z^{(n)}(\tau)}{Z(\tau)}.$$

Here P_n is not an additional fitted parameter. It is a deterministic derivative ratio generated by the node through its logarithmic derivative and its higher derivatives. The node value is handled in log-polar or multiplicative form, while derivative growth is handled through the normalized quotient P_n. This separates the large exponential or oscillatory lever from the derivative ratio and delays cancellation-prone real projection until the final aggregation step.

The contribution is representational and organizational. Novelty is not claimed for oscillations, complex exponentials, Bell polynomials, or ordinary differentiation taken separately; the contribution is the assembly of these standard objects into a disciplined local derivative calculus for machine-computation turning models, with explicit attention to cancellation, multiplicative normalization, and an optional imaginary-axis coordinate package. The imaginary-axis package is only a coordinate organization of the same real-coordinate derivatives unless a separate holomorphic extension is explicitly assumed. The claims are therefore structural and local, rather than claims of universal approximation or application-specific forecasting performance.

The numerical role of the framework is also limited in a specific way. A small path error does not by itself certify reliable first-, second-, or third-derivative behavior. The derivative symbols make these quantities available in closed form, but derivative quality still depends on conditioning, normalization, regularization, admissible evaluation sets, and the availability of a smooth reference when derivative-aware fitting is used. For this reason, the paper includes reference implementation checks and controlled synthetic stress tests as numerical verification of the calculus and fitting protocol, not as evidence of application-specific forecasting performance. This point is part of the model discipline rather than a separate empirical detail.

The Bell-polynomial and falling-factorial notation follows the classical exponential-polynomial and combinatorial literature [1, 2]. The local-window viewpoint is closest in spirit to local polynomial modeling, although the present object is derivative-symbol organization rather than nonparametric smoothing [3]. The oscillatory terminology is compatible with standard spectral and time-series language, but the representation here is finite and local rather than a global spectral model [7]. The cancellation and normalized-computation discussion follows the general numerical-stability concern that mathematically equivalent formulas can differ substantially in finite precision [4].

2 Local oscillatory-envelope model class

2.1 Single node

Fix a local horizon $T > 0$. A nonzero single oscillatory-envelope node on the positive local coordinate is defined by

$$Z(\tau) = b\tau^m e^{\alpha(\tau)} e^{\mathrm{i}(\omega\tau+\phi)}, \qquad 0 < \tau \le T, \tag{2}$$

where $b \in \mathbb{C} \setminus \{0\}$, $\omega \ge 0$, $\phi \in \mathbb{R}$, and α is real-valued and sufficiently smooth on an open neighborhood of the evaluation points under consideration. With these conventions, $Z(\tau) \ne 0$ for every $\tau > 0$. Zero-amplitude nodes are omitted because the quotient symbols $Z^{(n)}/Z$ are then undefined and the node contributes nothing to the signal. The nonnegative-frequency convention is not restrictive for the real-channel finite-sum model used below, since sign changes in frequency can be represented by the corresponding conjugate phase and coefficient choices before real projection.

For real m, the power τ^m is understood as the positive-real power on $\tau > 0$. No value at $\tau = 0$ is implied unless an explicit origin convention is imposed. In the main computational package, m is a small nonnegative integer, typically $m \in \{0, 1, 2\}$. A non-integer or decimal value of m is also mathematically meaningful on the positive coordinate $\tau > 0$, but then τ^m is treated as a fractional scale factor rather than as an ordinary polynomial term. Such a fractional-exponent extension is used either on $\tau \in [\tau_{\min}, T]$ with $\tau_{\min} > 0$ or with a shifted coordinate $\tau + \delta$ to avoid unstable derivative ratios at the origin. All derivative identities below are therefore read on an admissible evaluation set: either $0 < \tau \le T$ with no derivative evaluation exactly at the origin, $\tau \in [\tau_{\min}, T]$ for some $\tau_{\min} > 0$, or the shifted-coordinate formulation introduced in Section 9.

Definition 1 (Admissible evaluation set)**.** For a fixed derivative order n, an unshifted admissible evaluation set is any set $E \subset (0, T]$ such that every point of E has the derivatives of α required up to that order and satisfies $\tau > 0$. Equivalently, the unshifted formulas are ordinary pointwise identities on interior points of $(0, T)$ and extend to an endpoint T only when the relevant one-sided derivatives or a smooth local extension are specified. For a shifted node with parameter $\delta > 0$, a shifted admissible evaluation set may be a subset of $[0, T]$, because the scale ratio is $m/(\tau + \delta)$ and is well-defined at $\tau = 0$. When the derivative order and coordinate package are clear from context, the phrase "admissible evaluation set" means admissible for that order and package. All quotient symbols in this paper are evaluated only on the relevant admissible set. For a finite collection of nodes, the common admissible set is the intersection of the node-specific admissible sets after inactive zero-amplitude nodes have been removed.

Remark 1 (Endpoint convention for the zero scale)**.** When $m = 0$, the scale factor τ^m is used as the constant scale 1 whenever an endpoint value at $\tau = 0$ is explicitly adopted. Its logarithmic derivative contribution is then the absent term 0, not the literal undefined expression $0/\tau$ evaluated at the origin. For $m > 0$, unshifted quotient symbols are not evaluated at $\tau = 0$, even if some ordinary derivatives of the unshifted node may exist there. Endpoint quotient calculus is therefore handled either by the zero-scale convention with $m = 0$ or by the shifted package in Section 9.

Throughout the paper, an identity involving derivatives up to order n is read under the assumption that the corresponding envelope has the required regularity on an open real neighborhood of the admissible evaluation points, typically at least the derivatives needed by the displayed quotient symbols and product-rule recursion. This neighborhood-level condition is intentionally stronger than a bare pointwise derivative statement, because repeated product-rule and chain-rule identities require local differentiability information. Unless the imaginary-axis package is explicitly invoked, all derivatives are ordinary derivatives with respect to the real local coordinate τ.

The real cosine and sine channels are

$$X^{(c)}(\tau) = \operatorname{Re} Z(\tau), \tag{3}$$

$$X^{(s)}(\tau) = \operatorname{Im} Z(\tau), \tag{4}$$

when b is real. If b is complex, its argument may be absorbed into the phase or into the external coefficient of the node; this is a normalization choice and does not change the derivative symbols. More generally, a complex coefficient can be absorbed into amplitude and phase, and a real output can be written as

$$x(\tau) = a_0 + \operatorname{Re} \sum_{i=1}^{N} \gamma_i Z_i(\tau), \qquad \gamma_i \in \mathbb{C}. \tag{5}$$

Here $a_0 \in \mathbb{R}$ is the real offset, N is the number of active nodes in the finite sum, Z_i is the ith complex node, and γ_i is its complex real-channel coefficient before real projection. This form keeps the complex calculus compact while preserving a real-valued observed signal. In formulas that contain both an internal coefficient b_i and an external coefficient γ_i or Γ_i, only their product is identifiable from the real-channel signal unless an additional normalization is imposed. To avoid notational ambiguity, one may choose either of two equivalent normalizations: absorb constant complex phases into the coefficients and take the internal node amplitudes to be nonnegative real numbers, or keep complex internal amplitudes and use simpler external coefficients. The derivative symbols below are unaffected by any constant coefficient, constant phase, external real-channel coefficient, or constant envelope level; those quantities only rescale or rotate the node. The symbols depend only on the scale exponent, the nonconstant envelope derivatives, and the phase frequency. Thus coefficient normalization is an estimation convention, while derivative-symbol generation is a separate algebraic step.

2.2 Sparse constrained family

A purely unconstrained sum of nodes is too flexible for short-window turning analysis. The local model class is therefore restricted by four computational discipline conditions.

Definition 2 (Constrained local oscillatory-envelope family)**.** A sparse local oscillatory-envelope family on a short window consists of signals of the form

$$x(\tau) = a_0 + \mathrm{Re} \sum_{i=1}^{N} \gamma_i b_i \tau^{m_i} e^{\alpha_i(\tau)} e^{\mathrm{i}(\omega_i \tau + \phi_i)}, \tag{6}$$

Here $a_0 \in \mathbb{R}$ is the real intercept, $\gamma_i, b_i \in \mathbb{C}$ are constant coefficient factors, and $m_i, \alpha_i, \omega_i, \phi_i$ are node-specific scale, envelope, frequency, and phase parameters. Equivalently, one may write the effective coefficient as $c_i = \gamma_i b_i$ and use $c_i \tau^{m_i} e^{\alpha_i(\tau)} e^{\mathrm{i}(\omega_i \tau + \phi_i)}$. The separated notation $\gamma_i b_i$ is bookkeeping for internal node amplitude and external real-channel coefficient, not a claim of separate identifiability. The family is subject to

$$\sup_{0<\tau\leq T} |\alpha_i(\tau)| \leq \varepsilon_\alpha, \tag{7}$$

$$m_i \in \{0, 1, \ldots, m_{\max}\} \quad \text{in the integer package,} \tag{8}$$

$$0 \leq \omega_i \leq C_\omega / T, \tag{9}$$

$$N \leq N_{\max}, \tag{10}$$

where the node formula is used on $(0, T]$, ε_α, $m_{\max}$, C_ω, and $N_{\max}$ are small or moderate constants, and derivative evaluation at the origin is avoided unless a shifted coordinate has been introduced. Nodes with zero effective amplitude are removed before quotient symbols are formed. In this finite-sum convention, the constant complex phase of each node may equivalently be absorbed into the coefficient multiplying that node; the derivative identities are invariant under this normalization.

Condition (7) is understood after any constant envelope level has been absorbed into the node coefficient; it prevents the nonconstant envelope component from dominating the phase geometry. Condition (8) prevents the representation from becoming an ordinary high-order Taylor expansion and records the default integer-scale package. In the fractional-scale extension, (8) is replaced by $m_i \in [0, m_{\max}]$ together with the origin safeguards discussed below. Condition (9) prevents artificial high-frequency wiggles inside a short local window. Condition (10) forces the representation to retain only a small number of dominant local shape signals. For derivative-order work, these structural restrictions are supplemented by the required smoothness and boundedness of the envelope derivatives on the common admissible evaluation set. Thus small envelope level alone is not used as a substitute for derivative regularity. When estimation is discussed, the phrase "node coefficient" always means the effective constant coefficient multiplying the dynamic factor $\tau^{m_i}e^{\alpha_i(\tau)}e^{\mathrm{i}\omega_i\tau}$ after the chosen amplitude-phase normalization.

Remark 2 (Scope of the constrained family)**.** The restrictions in (7)–(10) are part of the paper-level model class. The derivative calculus below can be studied at the level of a single node, a shifted node, or a finite real-channel sum without restating every bounded-frequency, sparse-node, envelope-size, and integer-scale restriction each time. This separation is only a notational convenience. Any model described as belonging to the constrained local oscillatory-envelope family should still satisfy the full restrictions stated above, together with the required smoothness and admissibility conditions.

2.3 Smooth base and oscillatory perturbation

For computational testing, it is useful to separate smooth local behavior from oscillatory corrections. Write

$$Z(\tau) = Z_{\mathrm{base}}(\tau) + Z_{\mathrm{pert}}(\tau), \tag{11}$$

where

$$Z_{\mathrm{base}}(\tau) = \sum_{i=1}^{N_0} b_i^{(0)}\tau^{m_i^{(0)}}e^{\alpha_i^{(0)}(\tau)}e^{\mathrm{i}\phi_i^{(0)}}, \qquad \omega_i^{(0)} = 0, \tag{12}$$

and

$$Z_{\mathrm{pert}}(\tau) = \sum_{j=1}^{N_1} b_j^{(1)}\tau^{m_j^{(1)}}e^{\alpha_j^{(1)}(\tau)}e^{\mathrm{i}(\omega_j^{(1)}\tau+\phi_j^{(1)})}, \qquad \omega_j^{(1)} > 0. \tag{13}$$

Here Z_{base} denotes the smooth zero-frequency component and Z_{pert} denotes the oscillatory perturbation component. The parameters $b_i^{(0)}, m_i^{(0)}, \alpha_i^{(0)}, \phi_i^{(0)}$ are base-layer node parameters, while $b_j^{(1)}, m_j^{(1)}, \alpha_j^{(1)}, \omega_j^{(1)}, \phi_j^{(1)}$ are perturbation-layer node parameters. The perturbation layer is weak when N_1 is small, $|b_j^{(1)}|$ is controlled, and $\omega_j^{(1)}$ is bounded by the local scale. Here the superscript (0) labels smooth-base parameters, the superscript (1) labels perturbation-layer parameters, and N_0, N_1 denote the corresponding node counts.

This decomposition leads to a three-model comparison:

$$\mathrm{A}: \quad Z^{(A)}(\tau) = Z_{\mathrm{base}}(\tau), \tag{14}$$

$$\mathrm{B}: \quad Z^{(B)}(\tau) = Z_{\mathrm{base}}(\tau) + Z_{\mathrm{pert}}(\tau), \tag{15}$$

$$\mathrm{C}: \quad Z^{(C)}(\tau) = \sum_{i=1}^{N} b_i\tau^{m_i}e^{\alpha_i(\tau)}e^{\mathrm{i}(\omega_i\tau+\phi_i)}. \tag{16}$$

Model A tests a smooth local base, Model B tests whether weak oscillatory corrections add turning information, and Model C tests the broader constrained oscillatory family.

3 Factorized derivative calculus and cancellation control

3.1 Log-derivative symbol

For the single node (2), define the log-derivative symbol

$$q(\tau) = \frac{Z'(\tau)}{Z(\tau)} = \frac{m}{\tau} + \alpha'(\tau) + \mathrm{i}\omega. \tag{17}$$

For $j \geq 1$,

$$q^{(j)}(\tau) = (-1)^j j! \frac{m}{\tau^{j+1}} + \alpha^{(j+1)}(\tau). \tag{18}$$

Here $j = 0$ is not included in (18); the zeroth log-derivative symbol is q itself as defined in (17). The frequency ω appears in q but disappears from higher derivatives of q because the phase is linear in τ. Equation (18) is pointwise on the admissible set; it is not an instruction to evaluate the singular term at $\tau = 0$ in the unshifted model. If the zero-scale endpoint convention is used with $m = 0$, the scale contribution in q is replaced by 0 at the endpoint. Throughout this section, primes on q, P_n, and α denote ordinary derivatives with respect to the local coordinate τ.

Definition 3 (Derivative symbols)**.** For every derivative order for which the required derivatives of α exist, let $P_0(\tau) = 1$. Define recursively

$$P_{n+1}(\tau) = P_n'(\tau) + q(\tau)P_n(\tau), \qquad n \geq 0. \tag{19}$$

The function P_n is called the nth derivative symbol of the node.

The first four derivative symbols are

$$P_1 = q, \tag{20}$$

$$P_2 = q^2 + q', \tag{21}$$

$$P_3 = q^3 + 3qq' + q'', \tag{22}$$

$$P_4 = q^4 + 6q^2q' + 3(q')^2 + 4qq'' + q'''. \tag{23}$$

These are exactly the complete exponential Bell polynomials evaluated at $q, q', \ldots$, using the standard exponential-polynomial convention [1, 2]. In these displays, q, q', q'', q''' are all evaluated at the same admissible point τ; they are not independent variables. The useful algebraic check is not merely to introduce display abbreviations named $P_1, \ldots, P_4$, but to verify that the recursively generated symbols in (19) agree with these displayed formulas under the same admissibility and neighborhood-smoothness assumptions. To fix notation, let $Y_0 = 1$ and define the complete exponential Bell polynomials by

$$Y_{n+1}(x_1, \ldots, x_{n+1}) = \sum_{j=0}^{n} \binom{n}{j} Y_{n-j}(x_1, \ldots, x_{n-j})x_{j+1}, \qquad n \geq 0, \tag{24}$$

with the convention that Y_0 has no arguments. Then

$$Y_1 = x_1, \qquad Y_2 = x_1^2 + x_2, \qquad Y_3 = x_1^3 + 3x_1x_2 + x_3.$$

Theorem 1 (Factorized derivative theorem)**.** Fix a finite order $n \geq 0$. Let Z be the oscillatory-envelope node (2) with $b \neq 0$, with the positive-real convention for τ^m, and with zero-amplitude nodes removed before quotient symbols are formed. Fix an admissible evaluation point τ in the unshifted positive-coordinate package. Assume that α has the derivatives required by the order-n quotient-symbol recursion on a real neighborhood of that point, in particular the derivatives up to order n when P_n is written through $q, q', \ldots, q^{(n-1)}$, or has the corresponding one-sided derivatives if an endpoint convention has been explicitly declared. For the finite-order calculus used here, this is treated as a neighborhood-smoothness assumption, not merely as an isolated pointwise differentiability assertion. Then $Z(\tau) \neq 0$ and

$$Z^{(n)}(\tau) = Z(\tau)P_n(\tau), \tag{25}$$

where P_n is defined by (19). Equivalently,

$$P_n(\tau) = Y_n\big(q(\tau), q'(\tau), \dots, q^{(n-1)}(\tau)\big), \tag{26}$$

with the convention that the case $n = 0$ is the empty-argument identity $P_0 = Y_0 = 1$. Here Y_n denotes the complete exponential Bell polynomial. The quotient symbol P_n is therefore determined only by the logarithmic derivative layer $q, q', \dots, q^{(n-1)}$ and not by the absolute node scale. All quantities in this identity are evaluated at the same admissible point τ.

Proof. The case $n = 0$ is immediate. Assume $Z^{(n)} = ZP_n$. Since $Z' = Zq$,

$$Z^{(n+1)} = (ZP_n)' = Z'P_n + ZP_n' = Z(qP_n + P_n').$$

Therefore $P_{n+1} = P_n' + qP_n$. Induction gives (25) on every admissible point $\tau > 0$.

For the Bell-polynomial representation, work locally around the admissible evaluation point and choose a branch of the constant logarithm of b. Then

$$g(\tau) = \log b + m \log \tau + \alpha(\tau) + \mathrm{i}(\omega\tau + \phi)$$

satisfies $Z = e^g$ and $g' = q$. The chosen branch changes g only by a constant multiple of $2\pi\mathrm{i}$ and therefore does not change any derivative. Repeated differentiation of e^g gives the complete exponential Bell polynomials defined in (24), hence (26).

Remark 3 (Finite-order use)**.** Only finitely many derivative symbols are needed in the intended local turning package, typically P_1, P_2, and P_3. Therefore the Bell-polynomial notation is used as a finite-order organizational device, not as an asymptotic or infinite-series expansion.

Remark 4 (Finite-order verification convention)**.** The Bell-polynomial symbol Y_n in (24) is a recursively defined finite polynomial. In computations or symbolic checks, this polynomial family should be generated recursively, or the first few required symbols should be defined explicitly, rather than treated as an unexplained symbol. Likewise, the theorem above is a pointwise real-coordinate statement under the stated admissibility and regularity hypotheses; it is not a universal identity over all real inputs, over zero nodes, or at the unshifted origin. Derivative identities in this paper are read with explicit finite-order smoothness hypotheses on a neighborhood or within the relevant domain. In particular, repeated chain-rule and operator-transform relations are not asserted for arbitrary nonsmooth functions, and the first-four-symbol check means that the recursively generated P_n agrees with the displayed expression P_n^{display}. The ordinary neighborhood-derivative statements describe the interior positive-coordinate package. The optional one-sided endpoint convention requires a separate within-domain or one-sided derivative formulation and should not be inferred from the ordinary-neighborhood statements.

Remark 5 (Pointwise nature of the factorization)**.** The factorization is a node-level identity. It is valid at every admissible point where the node is nonzero. It does not require the real aggregate output to be nonzero, and it does not require division by a fitted real path.

3.2 Multiplicative quotient package

The factorization theorem suggests a machine-computation package based on multiplication and division rather than early additive expansion. The normalized derivative symbol

$$P_n(\tau) = \frac{Z^{(n)}(\tau)}{Z(\tau)} \tag{27}$$

contains the derivative information, while the node value $Z(\tau)$ contains the local amplitude and phase. Thus the implemented state of one node can be stored as

$$\mathcal{S}_n(\tau) = \big(\ell(\tau), \theta(\tau), P_0(\tau), P_1(\tau), \ldots, P_n(\tau)\big), \tag{28}$$

where

$$\ell(\tau) = \log|b| + m\log\tau + \alpha(\tau), \tag{29}$$

$$\theta(\tau) = \arg b + \omega\tau + \phi. \tag{30}$$

Here $\mathcal{S}_n$ denotes the finite node state stored up to derivative order n, ℓ is the log-amplitude state, and θ is the phase state. Any fixed branch of $\arg b$ may be used; changing the branch only adds an integer multiple of 2π to θ and leaves the reconstructed node and all quotient symbols unchanged. The complex node is reconstructed only when necessary as

$$Z(\tau) = e^{\ell(\tau)}\big(\cos\theta(\tau) + \mathrm{i}\sin\theta(\tau)\big). \tag{31}$$

This does not remove addition from the mathematics. The recursion

$$P_{n+1} = P_n' + qP_n$$

still contains additive terms. The numerical distinction is that the recursion is applied to derivative ratios rather than to expanded real components of $Z^{(n)}$ itself. In particular, powers such as $(\mathrm{i}\omega)^2 = -\omega^2$ and cross terms such as $2\mathrm{i}\omega\alpha'$ are kept inside a compact complex quotient until the real output is requested. This reduces unnecessary sign alternation inside each node. In implementation, the symbol $P_n = Z^{(n)}/Z$ should therefore be read primarily as a normalized analytic object. It need not be computed by numerically differentiating Z and then dividing by Z; it can be generated directly from the recursion or from the Bell-polynomial formula. This distinction avoids avoidable quotient noise when the node magnitude is small but nonzero, in line with the standard numerical principle that algebraic rearrangement can materially affect finite-precision stability [4].

From a scientific-computing viewpoint, the quotient package should not be implemented as a finite-difference estimate of $Z^{(n)}(\tau)/Z(\tau)$. Finite-difference derivatives introduce a step-size tradeoff between truncation error and roundoff error, and the difficulty becomes more severe for higher derivatives. The intended implementation instead forms P_n analytically from $q, q', \ldots, q^{(n-1)}$, using the recursion or the Bell-polynomial formula. The remaining numerical errors then come primarily from floating-point evaluation of the closed-form symbols, parameter estimation, and final real aggregation, rather than from numerical differentiation of the observed path.

Computational tests of this type should therefore separate identity verification from model validation. Recursion identities, scaled-coordinate identities, and imaginary-axis identities can be checked to machine precision for a fixed node. Fitted turning models require additional diagnostics: matrix conditioning, held-out path error, derivative error on a smooth reference, and sensitivity to the shifted-coordinate parameter. In particular, a level-only least-squares fit may have an acceptable path error while retaining a large second- or third-derivative error. Such a result indicates a fitting limitation, not a failure of the quotient identities.

A dimensionless formulation is obtained by setting $u = \tau/T$ with $T > 0$ and $u > 0$ in the unshifted positive-coordinate package. The symbol $\widetilde{P}_n$ is the derivative symbol for the rescaled function $u \mapsto Z(Tu)$, so it is an u-derivative symbol rather than another notation for the original τ-derivative symbol. Then

$$\widetilde{P}_n(u) = T^n P_n(Tu) = T^n \frac{Z^{(n)}(Tu)}{Z(Tu)}. \tag{32}$$

For the constant-envelope-slope case with $\lambda = r + \mathrm{i}\omega$ and $\mu = T\lambda$, the scaled symbol becomes

$$\widetilde{P}_n(u) = \sum_{k=0}^{n} \binom{n}{k} \frac{(m)_k}{u^k} \mu^{n-k}. \tag{33}$$

Here $(m)_0 = 1$ and $(m)_k = m(m-1)\cdots(m-k+1)$ for $k \geq 1$; this falling-factorial convention is stated again in Section 4 where the constant-slope law is proved. This form keeps the local horizon T out of the denominator and makes the computational levers explicit through the dimensionless quantities u^{-1} and μ. It is valid for the same positive-coordinate package as the unscaled symbol, so $T > 0$ and $u > 0$ are part of the dimensionless-coordinate convention unless the shifted-coordinate convention is used. It is also the quotient symbol associated with differentiating the rescaled function $u \mapsto Z(Tu)$, since

$$\frac{\mathrm{d}^n}{\mathrm{d}u^n} Z(Tu) = Z(Tu)\, \widetilde{P}_n(u).$$

If evaluation at $u = 0$ is required, the shifted-coordinate convention should be used.

Remark 6 (What multiplication and division can and cannot do)**.** A complete replacement of addition by multiplication and division is impossible for derivative algebra because differentiation of products necessarily creates sums. A more precise replacement is to avoid expanding complex exponentials into cancellation-prone real sums before the derivative ratios have been formed. The preferred computational primitive is the quotient package (Z, P_n), not the fully expanded real polynomial in sines and cosines.

3.3 Explicit first-, second-, and third-order node blocks

Substituting (17) and (18) into (20)-(22) gives the derivative blocks most relevant for turning analysis.

The first derivative is

$$Z'(\tau) = Z(\tau) \left[\alpha'(\tau) + \mathrm{i}\omega + \frac{m}{\tau}\right]. \tag{34}$$

The second derivative is

$$Z''(\tau) = Z(\tau) \left[(\alpha')^2 + \alpha'' + 2\mathrm{i}\omega\alpha' - \omega^2 + \frac{2m}{\tau}(\alpha' + \mathrm{i}\omega) + \frac{m(m-1)}{\tau^2}\right]. \tag{35}$$

The third derivative is

$$\begin{aligned} Z'''(\tau) = Z(\tau) \Bigg[& (\alpha')^3 + 3\alpha'\alpha'' + \alpha''' + 3\mathrm{i}\omega\big((\alpha')^2 + \alpha''\big) - 3\omega^2\alpha' - \mathrm{i}\omega^3 \\ & + \frac{3m}{\tau}\big((\alpha')^2 + \alpha'' + 2\mathrm{i}\omega\alpha' - \omega^2\big) + \frac{3m(m-1)}{\tau^2}(\alpha' + \mathrm{i}\omega) + \frac{m(m-1)(m-2)}{\tau^3} \Bigg]. \end{aligned} \tag{36}$$

All envelope derivatives in (35) and (36) are evaluated at τ.

Equations (34)-(36) make explicit the separation of three layers: an envelope derivative layer, a phase-frequency layer, and a polynomial-scale singular layer. The singular layer is structured by falling factorials rather than arbitrary coefficients.

4 Constant-envelope-slope specialization

A useful specialization is the constant-envelope-slope case, where $r \in \mathbb{R}$ and, on the admissible evaluation interval,

$$\alpha'(\tau) = r, \qquad \alpha''(\tau) = \alpha'''(\tau) = \cdots = 0. \tag{37}$$

Let

$$\lambda = r + \mathrm{i}\omega. \tag{38}$$

Then

$$Z(\tau) = \widetilde{b}\,\tau^m e^{\lambda\tau}, \tag{39}$$

where the constant phase and constant envelope level are absorbed into $\widetilde{b}$.

For an integer $k \geq 0$, define the falling factorial

$$(m)_0 = 1, \qquad (m)_k = m(m-1)\cdots(m-k+1), \quad k \geq 1. \tag{40}$$

This definition is algebraic in m and therefore remains meaningful for real, non-integer, or decimal m. For values of m where the gamma functions are finite, the same quantity can also be written as

$$(m)_k = \frac{\Gamma(m+1)}{\Gamma(m-k+1)}. \tag{41}$$

The product definition (40) is the primary convention in this paper; it also covers nonnegative integer m when the gamma quotient would otherwise require a limiting interpretation because the denominator meets a pole. Although the theorem below is algebraic for real m on $\tau > 0$, the constrained local model normally uses nonnegative m and prefers small integer values for stable computation. Negative values of m are not part of the default local-scale package because they introduce an additional pole at the origin.

Theorem 2 (Binomial falling-factorial derivative law)**.** Fix a finite order $n \geq 0$. For the constant-envelope-slope node (39), with $\widetilde{b} \neq 0$, $m \in \mathbb{R}$, and $\tau > 0$ under the positive-real power convention,

$$Z^{(n)}(\tau) = Z(\tau) \sum_{k=0}^{n} \binom{n}{k} \frac{(m)_k}{\tau^k} \lambda^{n-k}, \qquad n \geq 0. \tag{42}$$

For $n = 0$, the sum has only the term $k = 0$ and gives the identity $Z^{(0)} = Z$.

Proof. The node is $Z(\tau) = \widetilde{b}\,\tau^m e^{\lambda\tau}$. By the Leibniz rule,

$$Z^{(n)}(\tau) = \widetilde{b} \sum_{k=0}^{n} \binom{n}{k} \frac{\mathrm{d}^k}{\mathrm{d}\tau^k}\tau^m \frac{\mathrm{d}^{n-k}}{\mathrm{d}\tau^{n-k}} e^{\lambda\tau}.$$

For $\tau > 0$ and real m,

$$\frac{\mathrm{d}^k}{\mathrm{d}\tau^k}\tau^m = (m)_k \tau^{m-k}, \qquad \frac{\mathrm{d}^{n-k}}{\mathrm{d}\tau^{n-k}} e^{\lambda\tau} = \lambda^{n-k} e^{\lambda\tau}.$$

The first identity follows by induction from the positive-real convention for τ^m. Dividing by the nonzero quantity $Z(\tau) = \widetilde{b}\tau^m e^{\lambda\tau}$ yields (42).

Remark 7 (Non-integer and decimal scale exponents)**.** If m is a nonnegative integer, then $(m)_k = 0$ for every $k > m$. Hence the singular part of (42) terminates after finitely many polynomial-scale terms. For example, if $m = 2$, then $(m)_3 = 0$ and no τ^{-3} contribution is generated by the polynomial scale. If m is decimal, such as $m = 0.5$, the falling-factorial sequence does not terminate as the derivative order increases:

$$(0.5)_1 = 0.5, \qquad (0.5)_2 = -0.25, \qquad (0.5)_3 = 0.375.$$

Thus fractional scale exponents are allowed by the algebra, but they strengthen the need for an origin-excluding coordinate. The default package of this paper therefore keeps $m \in \{0, 1, 2\}$ unless a fractional-exponent extension is explicitly intended.

Remark 8 (Shifted fractional scale)**.** For machine computation with a decimal exponent, a safer node is

$$Z_\delta(\tau) = \widetilde{b}\,(\tau + \delta)^m e^{\lambda\tau}, \qquad \delta > 0. \tag{43}$$

The same Leibniz argument gives, for $\tau \geq 0$,

$$Z_\delta^{(n)}(\tau) = Z_\delta(\tau) \sum_{k=0}^{n} \binom{n}{k} \frac{(m)_k}{(\tau + \delta)^k} \lambda^{n-k}. \tag{44}$$

This formulation replaces m/τ by $m/(\tau + \delta)$ and avoids evaluating fractional-power derivative ratios at the current point. The condition $\delta > 0$ keeps the positive-real power $(\tau + \delta)^m$ single-valued without a complex branch choice.

For $n = 1, 2, 3$, formula (42) gives

$$Z'(\tau) = Z(\tau) \left[\lambda + \frac{m}{\tau}\right], \tag{45}$$

$$Z''(\tau) = Z(\tau) \left[\lambda^2 + \frac{2m}{\tau}\lambda + \frac{m(m-1)}{\tau^2}\right], \tag{46}$$

$$Z'''(\tau) = Z(\tau) \left[\lambda^3 + \frac{3m}{\tau}\lambda^2 + \frac{3m(m-1)}{\tau^2}\lambda + \frac{m(m-1)(m-2)}{\tau^3}\right]. \tag{47}$$

Thus the internal derivative system is not an unstructured collection of formulas. It is a binomial expansion in λ and τ^{-1}, with the polynomial-scale contribution governed by falling factorials.

Remark 9 (Interpretation of λ)**.** The real part r controls local envelope drift, while the imaginary part ω controls local oscillatory phase speed. In the derivative symbols, r and ω enter through the complex slope $\lambda = r + \mathrm{i}\omega$. This is one reason the constant-slope case is analytically compact.

5 Imaginary-axis coordinate package

A second way to organize the same algebra is to move the local coordinate from the positive real axis to the imaginary axis. This section is deliberately formulated as a contour-coordinate package. It should not be read as replacing the real-coordinate derivative engine by an unrestricted complex-holomorphic derivative unless a holomorphic extension is separately introduced and verified. Define

$$\zeta = \mathrm{i}\tau, \qquad \tau = -\mathrm{i}\zeta, \qquad \zeta \in \mathrm{i}(0, T]. \tag{48}$$

The purpose of this section is only to define an equivalent coordinate package. It is not necessary to assume that the envelope function α has a holomorphic extension away from the original real coordinate, and no contour integration is being introduced. The construction is a finite-order change of variables along the curve $\zeta = \mathrm{i}\tau$. To make this precise, if $F(\zeta) = f(-\mathrm{i}\zeta)$ is defined only on the contour $\mathrm{i}(0, T]$, define the contour derivative

$$\mathcal{D}_\zeta F(\zeta) := -\mathrm{i} f'(-\mathrm{i}\zeta). \tag{49}$$

When a holomorphic extension exists, $\mathcal{D}_\zeta$ agrees with the usual complex derivative on the contour. Otherwise, (49) is simply a change-of-coordinate derivative along the contour. In this paper, $\mathcal{D}_\zeta$ is interpreted as this contour-parametrized operator unless holomorphic-extension assumptions are added explicitly. Under this convention,

$$\frac{\mathrm{d}}{\mathrm{d}\tau} = \mathrm{i}\mathcal{D}_\zeta. \tag{50}$$

The imaginary-axis package is stated most directly for nonnegative integer m, so that no fractional-power branch is introduced. Along the contour, the node can be written as

$$\mathcal{Z}(\zeta) = b(-\mathrm{i}\zeta)^m e^{\alpha(-\mathrm{i}\zeta)} e^{\omega\zeta + \mathrm{i}\phi}. \tag{51}$$

All equalities in this section are contour equalities evaluated at $\zeta = \mathrm{i}\tau$. The oscillatory term $e^{\mathrm{i}\omega\tau}$ is now represented as $e^{\omega\zeta}$ on the imaginary contour. Since ζ remains imaginary, this is not a real-axis exponential blow-up; it is a coordinate rewrite of the same oscillation.

Define the imaginary-axis log-derivative symbol using the contour derivative:

$$q_\zeta(\zeta) = \frac{\mathcal{D}_\zeta \mathcal{Z}(\zeta)}{\mathcal{Z}(\zeta)} = \frac{m}{\zeta} - \mathrm{i}\alpha'(-\mathrm{i}\zeta) + \omega. \tag{52}$$

In this formula, $\alpha'(-\mathrm{i}\zeta)$ means the ordinary real-coordinate derivative of α evaluated at the real point $\tau = -\mathrm{i}\zeta$ on the contour; it is not a separate complex derivative assumption. When $\zeta = \mathrm{i}\tau$, the argument $-\mathrm{i}\zeta$ is the original real coordinate τ, so no off-contour value of α is used. Therefore, on the contour,

$$q_\zeta(\mathrm{i}\tau) = -\mathrm{i}q(\tau), \qquad q(\tau) = \mathrm{i}q_\zeta(\mathrm{i}\tau), \tag{53}$$

where $q(\tau) = m/\tau + \alpha'(\tau) + \mathrm{i}\omega$ is the original real-axis log-derivative symbol. This sign convention follows from the contour derivative $\mathcal{D}_\zeta = -\mathrm{i}\,\mathrm{d}/\mathrm{d}\tau$ along $\zeta = \mathrm{i}\tau$.

Proposition 1 (Coordinate equivalence)**.** Assume the admissibility, integer-scale or branch, and finite-order differentiability conventions stated above for the contour point $\zeta = \mathrm{i}\tau$. In addition, assume the pulled-back contour function is differentiable to the required finite order along the contour; equivalently, the following statement is not made for an arbitrary nonsmooth contour function. Let $\mathcal{P}_0 = 1$ and

$$\mathcal{P}_{n+1} = \mathcal{D}_\zeta \mathcal{P}_n + q_\zeta \mathcal{P}_n, \tag{54}$$

where $\mathcal{D}_\zeta$ is the contour derivative. Then

$$Z^{(n)}(\tau) = \mathrm{i}^n \mathcal{D}_\zeta^n \mathcal{Z}(\mathrm{i}\tau) = Z(\tau)\, \mathrm{i}^n \mathcal{P}_n(\mathrm{i}\tau). \tag{55}$$

Therefore

$$P_n(\tau) = \mathrm{i}^n \mathcal{P}_n(\mathrm{i}\tau), \qquad \mathcal{P}_n(\mathrm{i}\tau) = (-\mathrm{i})^n P_n(\tau). \tag{56}$$

Proof. Under the finite-order contour smoothness assumption, the operator identity (50) gives $Z^{(n)}(\tau) = \mathrm{i}^n \mathcal{D}_\zeta^n \mathcal{Z}(\mathrm{i}\tau)$. Applying the same quotient-symbol proof to $\mathcal{Z}$ with respect to the contour derivative $\mathcal{D}_\zeta$ gives $\mathcal{D}_\zeta^n \mathcal{Z} = \mathcal{Z}\mathcal{P}_n$. Since $\mathcal{Z}(\mathrm{i}\tau) = Z(\tau)$, equation (55) follows.

The imaginary-axis package is therefore not a different model. It is a coordinate package for the same derivative engine, with equivalence understood at the same finite derivative order and on the same admissible evaluation set. The identity $\mathcal{Z}(\mathrm{i}\tau) = Z(\tau)$ follows simply from $-\mathrm{i}(\mathrm{i}\tau) = \tau$; no value away from the original contour is needed. It is useful when an implementation wants to treat oscillation as motion along an imaginary contour while keeping exponential notation multiplicative. The notation $\alpha(-\mathrm{i}\zeta)$ in this section means the original real-coordinate envelope evaluated at $\tau = -\mathrm{i}\zeta$ along the contour; it does not by itself assert a holomorphic extension of α. If m is not an integer, the factor $(-\mathrm{i}\zeta)^m$ requires a branch convention if it is treated as a complex power. For this reason, practical implementations should either use integer m, use the shifted positive scale introduced in Section 9, or treat the imaginary-axis form only as a phase-coordinate layer while evaluating the scale factor as the positive-real quantity τ^m or $(\tau + \delta)^m$ in the original coordinate.

Remark 10 (Contour-pullback convention)**.** To avoid off-contour complex powers, the imaginary-axis package may be read through the contour pullback $\tau \mapsto Z(\tau)$ together with the contour operator $\mathcal{D}_\zeta = -\mathrm{i}\,\mathrm{d}/\mathrm{d}\tau$. This convention establishes the coordinate equivalence in (53)–(56) directly on the contour, without requiring a holomorphic extension of α or a global branch of $(-\mathrm{i}\zeta)^m$. The pullback convention is faithful to the derivative-symbol equivalence on $\zeta = \mathrm{i}\tau$, but it is not the same as imposing a full off-contour complex-coordinate model from (51). A direct off-contour reading of (51) would require additional branch and holomorphic-extension assumptions.

6 Real-channel derivative blocks

Let the real local signal be a finite sum of nonzero smooth nodes,

$$x(\tau) = a_0 + \mathrm{Re}\sum_{i=1}^{N} \Gamma_i Z_i(\tau), \qquad \Gamma_i \in \mathbb{C}. \tag{57}$$

Here $a_0 \in \mathbb{R}$ is the real offset, N is the number of active nodes, and Γ_i is the complex external coefficient multiplying the ith node before real projection. For each node define

$$q_i(\tau) = \frac{m_i}{\tau} + \alpha_i'(\tau) + \mathrm{i}\omega_i, \tag{58}$$

and let $P_{n,i}$ be generated by

$$P_{0,i} = 1, \qquad P_{n+1,i} = P_{n,i}' + q_i P_{n,i}. \tag{59}$$

Proposition 2 (Real-channel derivative formula)**.** Fix a finite order $n \geq 0$. At every point in the common admissible set where all active nodes are admissible for derivative order n and satisfy the corresponding neighborhood-smoothness assumptions, where active means that the node has nonzero effective coefficient and nonzero amplitude and inactive nodes have been omitted from the finite sum, the zero-th order identity is

$$x(\tau) = a_0 + \mathrm{Re}\sum_{i=1}^{N} \Gamma_i Z_i(\tau) P_{0,i}(\tau), \qquad P_{0,i} = 1, \tag{60}$$

and for every $n \geq 1$,

$$x^{(n)}(\tau) = \operatorname{Re} \sum_{i=1}^{N} \Gamma_i Z_i(\tau) P_{n,i}(\tau). \tag{61}$$

If a shifted-coordinate scale is used for a node, the same formula holds with that node's shifted symbol $P_{n,i,\delta}$ in place of $P_{n,i}$.

Proof. The case $n = 0$ is exactly (57) with $P_{0,i} = 1$. For $n \geq 1$, the constant a_0 differentiates to zero. The node-level factorization gives $Z_i^{(n)} = Z_i P_{n,i}$ for each active node. The sum in (57) is finite, so differentiation passes through the sum without any convergence issue. Since real projection is a real-linear operation and the coordinate derivative is taken componentwise on complex-valued functions, $\frac{\mathrm{d}^n}{\mathrm{d}\tau^n} \operatorname{Re} F = \operatorname{Re} F^{(n)}$ for each smooth complex-valued function F. Applying these facts to (57) gives (61).

Formula (61) is the practical derivative engine for orders $n \geq 1$, together with (60) for the signal itself. Once the node values and derivative symbols are computed, the real-channel derivatives are obtained by summation over the finite active node set. Complex-valued node derivatives are assembled first, and the real projection is taken only at the final aggregation stage. The quotient symbols are formed node by node; the construction never requires division by the real aggregate signal $x(\tau)$. Hence zero crossings of the real output do not create quotient singularities, although individual zero-amplitude nodes must still be removed before forming $P_{n,i}$.

6.1 Minimal sinusoidal submodel

A minimal real submodel is

$$x(\tau) = a + b_1 \cos(\omega\tau) + b_2 \sin(\omega\tau). \tag{62}$$

Here $a, b_1, b_2 \in \mathbb{R}$ are real coefficients and $\omega \geq 0$ is the fixed local frequency. The derivatives are

$$x'(\tau) = -b_1\omega \sin(\omega\tau) + b_2\omega \cos(\omega\tau), \tag{63}$$

$$x''(\tau) = -b_1\omega^2 \cos(\omega\tau) - b_2\omega^2 \sin(\omega\tau), \tag{64}$$

$$x'''(\tau) = b_1\omega^3 \sin(\omega\tau) - b_2\omega^3 \cos(\omega\tau). \tag{65}$$

This submodel has no envelope and no polynomial-scale singularity. It therefore provides a baseline for the pure phase contribution to local derivatives.

7 Turning-point definitions and indicators

The word "turning" can refer to different local events. A precise paper-level formulation separates at least two cases.

Definition 4 (Local extremum-type turn)**.** A point $\tau_* \in (0, T)$ is an extremum-type turn of a smooth real signal x if

$$x'(\tau_*) = 0, \qquad x''(\tau_*) \neq 0. \tag{66}$$

The sign of $x''(\tau_*)$ distinguishes local minima from local maxima.

Definition 5 (Inflection-type bend transition)**.** A point $\tau_* \in (0,T)$ is an inflection-type bend transition if

$$x''(\tau_*) = 0, \qquad x'''(\tau_*) \neq 0. \tag{67}$$

This event marks a change in curvature rather than necessarily a local path reversal.

These two cases are distinct and should not be conflated. A local maximum or minimum is a directional reversal. An inflection-type transition is a bend-geometry change. In any noisy path data, both may be relevant, but their derivative signatures differ.

7.1 Derivative-based local scores

Let $s_1, s_2, s_3 > 0$ be fixed positive scale constants estimated from local derivative magnitudes or chosen from a reference benchmark. Positivity is required so that the following normalized scores are well-defined. Two example dimensionless scores are

$$R_{\mathrm{ext}}(\tau) = \exp\left(-\frac{x'(\tau)^2}{2s_1^2}\right)\frac{|x''(\tau)|}{s_2 + |x''(\tau)|}, \tag{68}$$

and

$$R_{\mathrm{infl}}(\tau) = \exp\left(-\frac{x''(\tau)^2}{2s_2^2}\right)\frac{|x'''(\tau)|}{s_3 + |x'''(\tau)|}. \tag{69}$$

The first score is large when direction is close to zero and curvature is nontrivial. The second score is large when curvature is close to zero but curvature momentum is nontrivial.

These formulas are not required for the factorized calculus itself. They are one possible way to convert derivative blocks into local turning features. Other monotone transformations can be substituted without changing the derivative engine.

In grid-based computation, candidate rows should usually be clustered into local intervals before being counted as turning events. Adjacent rows often represent the same sign change or the same small-neighborhood curvature event. Reporting a representative point, an interval width, and a region score is more stable than reporting every flagged grid point as a separate turn. This is a numerical reporting convention and does not alter the definitions above.

8 Fitting and computational validation protocol

8.1 Local fitting problem

Given observations (τ_k, y_k), $k = 1, \ldots, K_y$, on a local window, a constrained oscillatory-envelope fit solves

$$\widehat{\vartheta} = \arg\min\nolimits_{\vartheta\in\Theta}\left\{\frac{1}{K_y}\sum_{k=1}^{K_y}\big(\widehat{x}_\vartheta(\tau_k) - y_k\big)^2 + \mathcal{R}(\vartheta)\right\}, \tag{70}$$

where ϑ denotes the fitted parameter vector, Θ encodes the sparsity, envelope, degree, and frequency restrictions, and $\mathcal{R}$ is a regularization term. The notation K_y is used here to distinguish the observation grid from any derivative-evaluation grid introduced later. A typical regularizer may penalize excessive envelope slope, excessive frequency, or large perturbation amplitude.

When nonlinear parameters such as ω_i and envelope parameters are fixed, the real-channel coefficients are estimated by linear least squares. This separation allows a practical two-level procedure: grid or constrained nonlinear search for structural parameters, followed by linear coefficient estimation. The fitting problem determines a smooth representation; derivative formulas are then evaluated from that fitted representation on its admissible set rather than obtained by differentiating raw discrete observations. Thus the derivatives used below are model-implied derivatives, not direct finite-difference derivatives of the observed sequence.

The nonlinear part of the fitting problem should be treated as a constrained numerical optimization problem. In practical computation, structural parameters such as frequencies, envelope parameters, and shift parameters should be scaled before optimization, and the coefficient-estimation subproblem should be solved separately when it is linear. Line-search, quasi-Newton, or safeguarded grid procedures may be used for the nonlinear layer, following standard numerical-optimization practice, but comparison should be made through out-of-sample or held-out-window diagnostics rather than only through in-sample path error [6].

The linear coefficient subproblem should also report conditioning. Purely inactive real or imaginary columns should be removed before least-squares estimation, and the remaining columns should be normalized when possible, in line with standard numerical linear algebra practice for least-squares problems [5]. When derivative rows are added to the least-squares system, the stacked matrix should be scaled by the derivative magnitudes and the chosen weights. This scaling can improve numerical conditioning and makes derivative-aware comparisons meaningful across derivative orders.

8.2 Derivative-aware objective

A turning model is not adequately evaluated by ordinary path fit alone. Let x denote a target path or a smoothed reference path, and let $\widehat{x}$ denote the local fitted signal. On the observation grid τ_k, $k = 1, \ldots, K_y$, with $K_y > 0$, define

$$L_{\text{fit}} = \frac{1}{K_y} \sum_{k=1}^{K_y} \big(\widehat{x}(\tau_k) - x(\tau_k)\big)^2. \tag{71}$$

Let τ_* be a target turning time and $\widehat{\tau}_*$ a predicted turning time. Both should be interpreted on the same local window, and in derivative-based implementations the search for $\widehat{\tau}_*$ should be restricted to the common admissible evaluation set. Define

$$L_{\text{turn}} = |\widehat{\tau}_* - \tau_*|. \tag{72}$$

For derivative evaluation, choose a grid ξ_j, $j = 1, \ldots, K_d$, with $K_d > 0$, contained in the common admissible evaluation set. This derivative grid may equal the observation grid, but it need not. Since derivative magnitudes can increase rapidly with order, the derivative-shape criterion should be scaled before different orders are combined. Let $s_1, s_2, s_3 > 0$ be reference scales, for example root-mean-square or robust quantile scales of the corresponding smooth reference derivatives, and let $\lambda_1, \lambda_2, \lambda_3 \geq 0$ be externally chosen derivative weights. A normalized derivative-shape loss is

$$L_{\text{shape}} = \sum_{\ell=1}^{3} \lambda_\ell \frac{1}{K_d} \sum_{j=1}^{K_d} \left(\frac{\widehat{x}^{(\ell)}(\xi_j) - x^{(\ell)}(\xi_j)}{s_\ell} \right)^2. \tag{73}$$

The combined evaluation criterion is

$$L = L_{\text{fit}} + \eta L_{\text{turn}} + L_{\text{shape}}. \tag{74}$$

Here $\eta \geq 0$ and $\lambda_\ell \geq 0$ are externally chosen weights; setting one of them to zero removes the corresponding component from the diagnostic criterion. The normalization by s_ℓ is part of the numerical specification. Without this normalization, high-order derivatives may dominate the objective only because they are measured on a larger numerical scale.

For noisy computational observations, x', x'', and x''' should be interpreted as derivatives of a smoothed target, synthetic benchmark, or validated reference curve rather than raw finite differences of noisy observations. The derivative identities themselves apply to the fitted smooth representation, not to a nonsmooth observed sequence. If no reliable derivative reference is available, L_{shape} should be treated as a diagnostic or omitted from the fitted objective rather than estimated from unstable high-order finite differences.

When a smooth derivative reference is available, derivative-aware fitting can be implemented as a scaled stacked least-squares problem. For fixed nonlinear structure, let $A^{(\ell)}$ be the real-channel design matrix whose columns evaluate the ℓth derivatives of the active nodes, with $A^{(0)}$ the ordinary level design matrix. Let $d^{(\ell)}$ denote the smooth reference derivative vector on the derivative grid, let $s_0, s_1, s_2, s_3 > 0$ be the level and derivative scaling constants, and let $\lambda_1, \lambda_2, \lambda_3, \rho \geq 0$ be the derivative and ridge weights. A representative coefficient-estimation problem is

$$\widehat{\beta} = \arg\min_\beta \left\{ \left\| \frac{A^{(0)}\beta - y}{s_0} \right\|_2^2 + \sum_{\ell=1}^{3} \lambda_\ell \left\| \frac{A^{(\ell)}\beta - d^{(\ell)}}{s_\ell} \right\|_2^2 + \rho \|\beta\|_2^2 \right\}. \tag{75}$$

Here $y = (y_1, \ldots, y_{K_y})$ is the observation vector, $\widehat{\beta}$ is the fitted coefficient vector, and β is the generic coefficient vector over which the least-squares objective is minimized. This equation is not a new derivative identity. It is the numerical fitting layer suggested by the derivative diagnostics. Its purpose is to prevent a purely level-based fit from hiding large derivative errors. The derivative targets in (75) should come from a smooth reference or controlled benchmark; they should not be produced by high-order finite differences of noisy observations.

8.3 A/B/C computational comparison

The base-perturbation setup in (14)-(16) yields the following validation logic.

Model	Structure	Computational question
A	Smooth base, $\omega = 0$	Is smooth local geometry sufficient?
B	Smooth base plus weak oscillatory perturbation	Does mild phase correction improve turns?
C	Full constrained oscillatory-envelope family	Is the local region strongly oscillatory?

If Model B improves over Model A in L_{turn} or L_{shape} without excessive deterioration in L_{fit}, the weak oscillatory perturbation contains useful local turning information. If Model C materially outperforms both, the local region may require a broader oscillatory representation. If Model B and Model C fail to improve over Model A, the smooth local base is the more defensible specification for that window.

8.4 Reference implementation and synthetic stress tests

A reference Python implementation, maintained at `https://github.com/DavidHShen/FDC_for_Local_Turning`, was used to verify the derivative-symbol calculus and the fitting protocol. The implementation forms the normalized symbols $P_n = Z^{(n)}/Z$ analytically from $q, q', \ldots, q^{(n-1)}$, rather than estimating derivative ratios by finite differences. Finite differences are used only as an external validation check. The implementation also uses the shifted-coordinate safeguard near the origin, delays real-channel aggregation until after node-level complex derivatives are formed, removes numerically inactive least-squares columns, reports condition numbers, and clusters adjacent turning candidates into intervals.

The first implementation check compares the recursive construction of P_n with the closed-form binomial expression in the constant-envelope-slope case. In double precision, the maximum relative error through order six is approximately 4.75×10^{-15}. The dimensionless-coordinate check gives a maximum relative error of approximately 4.67×10^{-16} through order four, and the imaginary-axis equivalence check is exact to the reported precision through order four. A centered finite-difference validation gives maximum relative errors of approximately 6.70×10^{-13}, 6.47×10^{-9}, and 2.44×10^{-6} for the first, second, and third derivatives, respectively. These results are consistent with the intended use of finite differences as validation tools rather than as the derivative engine.

Model	Cond.	Holdout x	Holdout x'	Holdout x''	Holdout x'''
Base only	173.11	0.43967	1.82270	9.01517	146.33771
Level-only oscillatory	932.82	0.07327	0.40756	4.44994	49.59362
Derivative-aware oracle	102.71	0.00155	0.00033	0.00441	0.03128

In the table, "Cond." denotes the condition number of the fitted least-squares design matrix, and the holdout columns report held-out errors for the level signal x and its first three derivatives.

The table reports results from a controlled synthetic benchmark in which a known smooth reference model is observed with additive Gaussian noise. The level-only oscillatory fit improves the path fit relative to the smooth base, but its higher-derivative errors remain substantial. The derivative-aware oracle fit improves both conditioning and derivative recovery because the stacked least-squares system includes smooth reference derivative targets. These targets are the analytic first-, second-, and third-derivative values $d^{(1)}, d^{(2)}, d^{(3)}$ generated from the known synthetic reference model on the training grid. This experiment is therefore consistent with the distinction between path accuracy and derivative accuracy.

A separate high-noise stress test examines the same fitting protocol for observation-noise scales $0.01, 0.03, 0.06, 0.10$, and 0.15. At the highest noise scale, the level-only oscillatory fit has holdout path error about 1.03138 and holdout third-derivative error about 664.06937, while the derivative-aware oracle fit has corresponding errors about 0.00397 and 0.80018. This stress test suggests that derivative-aware supervision can stabilize the fitted derivative structure in this controlled benchmark. It should not be interpreted as real-market predictive validation, because the derivative-aware case uses analytic derivative targets from the known synthetic reference model. In applications where such reference derivatives are not available, derivative quantities should remain diagnostics unless they are obtained from a defensible smoothing, calibration, or model-implied procedure.

9 Regularization, identifiability, and the origin problem

9.1 Origin singularity

The factor τ^m creates the log-derivative term m/τ. This term is singular at $\tau = 0$ when $m \neq 0$. The issue is sharper for non-integer or decimal m: although τ^m is well-defined for $\tau > 0$, it does not behave like an ordinary polynomial at the origin, and the falling-factorial coefficients $(m)_k$ do not automatically vanish for $k > m$. Higher derivative ratios can therefore contain persistent powers $\tau^{-1}, \tau^{-2}, \tau^{-3}, \ldots$ as the derivative order increases. Since $\tau = 0$ is the current reference time, the model treats the origin explicitly.

Three standard treatments are available.

(i) Restrict derivative evaluation to $\tau \in [\tau_{\min}, T]$ with $\tau_{\min} > 0$.

(ii) Use $m = 0$ for nodes evaluated at the current point, interpreting the scale as the constant 1 and the scale log-derivative contribution as absent rather than as a literal $0/0$ term. Allow $m > 0$ only away from the origin.

(iii) Replace τ^m by a shifted local scale

$$\left(\frac{\tau + \delta}{T + \delta}\right)^m, \qquad \delta > 0. \tag{76}$$

Then the log-derivative contribution becomes $m/(\tau + \delta)$ rather than m/τ.

The shifted-coordinate formulation is often the most convenient numerical choice because it preserves the polynomial-scale interpretation while avoiding a singular derivative at the current time. It is especially important when m is non-integer, because the generalized falling-factorial terms then do not terminate automatically. The distinction is between ordinary endpoint derivatives of a fitted expression and normalized quotient symbols: a polynomial node may have some ordinary derivatives at $\tau = 0$, but the quotient ratio $Z^{(n)}/Z$ is not used at an endpoint where the node itself vanishes.

The shift parameter δ should not be treated as a harmless cosmetic constant. The shifted symbols are finite at $\tau = 0$, but higher-order symbols can still be large when δ is too small. For a fixed maximum derivative order $n_{\max}$ and a small cutoff $\tau_0 \in [0, T]$ near the origin, an implementation should monitor quantities such as $\max_{0 \le \tau \le \tau_0} |P_{\ell,\delta}(\tau)|$ for $1 \le \ell \le n_{\max}$ and increase δ, lower the derivative order, or exclude the origin region when these quantities are numerically excessive. Thus the shifted coordinate removes the singularity, but it does not remove the need for order-aware conditioning checks.

For the general shifted node

$$Z_\delta(\tau) = b\left(\frac{\tau + \delta}{T + \delta}\right)^m e^{\alpha(\tau)} e^{\mathrm{i}(\omega\tau + \phi)}, \qquad \delta > 0, \tag{77}$$

the log-derivative symbol becomes

$$q_\delta(\tau) = \frac{m}{\tau + \delta} + \alpha'(\tau) + \mathrm{i}\omega, \tag{78}$$

with higher derivatives

$$q_\delta^{(j)}(\tau) = (-1)^j j! \frac{m}{(\tau + \delta)^{j+1}} + \alpha^{(j+1)}(\tau), \qquad j \ge 1. \tag{79}$$

Here $j = 0$ is not included in (79); the zeroth symbol is q_δ itself as defined in (78).

Proposition 3 (Shifted factorized derivative package)**.** Fix a finite order $n \geq 0$. Let Z_δ be defined by (77), with $b \neq 0$, $T > 0$, $\delta > 0$, and with α sufficiently smooth for the derivative order under consideration on a neighborhood of the relevant shifted admissible set. Define $P_{0,\delta} = 1$ and

$$P_{n+1,\delta} = P'_{n,\delta} + q_\delta P_{n,\delta}. \tag{80}$$

Then, for every $\tau \in [0, T]$ at which the required derivatives exist,

$$Z_\delta^{(n)}(\tau) = Z_\delta(\tau) P_{n,\delta}(\tau), \tag{81}$$

and

$$P_{n,\delta}(\tau) = Y_n\big(q_\delta(\tau), q'_\delta(\tau), \ldots, q_\delta^{(n-1)}(\tau)\big). \tag{82}$$

For $n = 0$, this is again read as the empty-argument identity $P_{0,\delta} = Y_0 = 1$.

Proof. The proof is identical to the proof of the factorized derivative theorem after replacing the logarithmic scale $m \log \tau$ by $m \log(\tau + \delta) - m \log(T + \delta)$. Since $\delta > 0$, this logarithm is real-valued on $[0, T]$, and the constant term $-m \log(T + \delta)$ disappears from all quotient symbols.

Thus the shifted-coordinate scale does not change the derivative calculus; it only replaces the singular ratio m/τ by the bounded ratio $m/(\tau + \delta)$ on $[0, T]$. In particular, the scale contribution is bounded in magnitude by $|m|/\delta$ on the full closed window. The normalization factor $(T + \delta)^{-m}$ is constant and therefore does not affect quotient symbols. If non-integer m is used in the shifted-coordinate scale, the positive condition $\tau + \delta > 0$ fixes the real branch of the power.

Proposition 4 (Shifted constant-slope binomial law)**.** Assume in (77) that $b \neq 0$, $T > 0$, $\delta > 0$, and $\alpha'(\tau) = r$ is constant on the shifted admissible set, and set $\lambda = r + \mathrm{i}\omega$. Then, for every finite order $n \geq 0$ and every $\tau \in [0, T]$ at which the required endpoint or interior derivative is defined,

$$Z_\delta^{(n)}(\tau) = Z_\delta(\tau) \sum_{k=0}^{n} \binom{n}{k} \frac{(m)_k}{(\tau + \delta)^k} \lambda^{n-k}. \tag{83}$$

Equivalently, the shifted quotient symbol is

$$P_{n,\delta}(\tau) = \sum_{k=0}^{n} \binom{n}{k} \frac{(m)_k}{(\tau + \delta)^k} \lambda^{n-k}. \tag{84}$$

For $n = 0$, the quotient symbol is again $P_{0,\delta} = 1$.

Proof. Since $\alpha'(\tau) = r$ is constant on the evaluation window, write $\alpha(\tau) = r\tau + c$ for a constant c. Then the shifted node can be written as

$$Z_\delta(\tau) = C(\tau + \delta)^m e^{\lambda\tau}, \qquad C = b(T + \delta)^{-m} e^{c + \mathrm{i}\phi},$$

where C is independent of τ and $\lambda = r + \mathrm{i}\omega$. Applying the Leibniz rule to $(\tau + \delta)^m e^{\lambda\tau}$ gives (83); division by the nonzero node gives (84).

9.2 Identifiability

Complex oscillatory nodes are not uniquely parameterized. Amplitude, phase, and real-channel coefficients can trade off against one another. This does not affect the derivative calculus, but it matters for statistical estimation.

After absorbing complex phases into the coefficients Γ_i, a practical specification can impose the following normalizations:

$$b_i \geq 0, \tag{85}$$

$$\phi_i \in (-\pi, \pi], \tag{86}$$

$$0 \leq \omega_1 \leq \omega_2 \leq \cdots \leq \omega_N, \tag{87}$$

$$\|Z_i\|_{L^2(0,T)} = 1 \quad \text{before coefficient estimation.} \tag{88}$$

Here $\|Z_i\|_{L^2(0,T)}$ denotes the usual square-integral norm of the node over the local window before coefficient estimation. These restrictions do not change the finite real span of the model class after coefficients are re-estimated, but they reduce redundant representations. In particular, they should be viewed as estimation normalizations rather than additional derivative assumptions.

9.3 Overfitting control

The constrained model is intended for local turning analysis rather than universal approximation. Overfitting control should therefore be structural. Relevant controls include small node count, bounded frequency, mild envelope slope, small polynomial order, and perturbation penalties in Model B. A representative penalty is

$$\mathcal{R}(\vartheta) = \rho_N N + \rho_\omega \sum_i (\omega_i T)^2 + \rho_\alpha \sum_i \|\alpha_i'\|^2_{L^2(0,T)} + \rho_b \sum_{j \in \text{pert}} |b_j^{(1)}|^2. \tag{89}$$

Here $\rho_N, \rho_\omega, \rho_\alpha, \rho_b \geq 0$, and pert is the index set of perturbation nodes in the base-plus-perturbation model. The norm $\|\alpha_i'\|^2_{L^2(0,T)}$ denotes the integrated squared envelope-slope magnitude on the local window. The penalty should be chosen to support local derivative stability, not merely in-sample path accuracy.

10 Relation to Taylor and Fourier viewpoints

A local Taylor approximation emphasizes polynomial behavior around a point, and local polynomial modeling provides a standard reference point for this style of local representation [3]. A Fourier or spectral representation emphasizes sinusoidal components, often in a more global or periodic setting [7]. The oscillatory-envelope model sits between these two viewpoints. The polynomial scale τ^m allows local algebraic behavior, the envelope $e^{\alpha(\tau)}$ allows mild local deformation, and the phase factor $e^{\mathrm{i}\omega\tau}$ allows controlled bend geometry.

The factorized derivative calculus is the main structural distinction. The model is not introduced only to fit $x(\tau)$; it is introduced because $x'(\tau)$, $x''(\tau)$, and $x'''(\tau)$ are available in closed structured form. This makes the representation potentially suitable for derivative-sensitive local tasks, including reversal screening, curvature-aware feature extraction, turn-time localization, and pre-filtering before heavier computational models.

11 Limitations and scope

The framework has several limitations.

First, the model is local. It should not be interpreted as a global law for the underlying path. Parameters estimated on one short window need not transfer across regimes or computational domains.

Second, derivative quality is harder to validate than path fit. In noisy data, second and third derivatives require smoothing, synthetic benchmarks, or carefully designed reference estimates. A level-only fit can appear accurate in path space while remaining poor in derivative space; this is why derivative-aware diagnostics and, when justified, derivative-aware fitting are part of the computational protocol.

Third, oscillatory perturbations can overfit. The bounded-frequency and sparse-node restrictions are therefore part of the model definition rather than optional implementation details.

Fourth, factorization provides a derivative representation, not a direct application guarantee. Any downstream application requires separate stability, conditioning, and out-of-sample evaluation.

Fifth, the derivative-symbol framework is used only for fixed finite derivative orders and under the stated admissibility, nonzero-node, branch, endpoint, and smoothness assumptions. The Bell-polynomial recursion should be read as the mechanism generating the derivative symbols, not as an opaque label. The displayed formulas for $P_1, \ldots, P_4$ are meant to agree with the recursively generated symbols. The unshifted dimensionless coordinate package requires $T > 0$ and $u > 0$, derivative losses and stacked least-squares objectives require positive normalization scales and nonnegative weights, and the imaginary-axis derivative is the contour operator (49) unless a holomorphic extension is separately assumed. Chain-rule and repeated-derivative operator transformations are not universal identities over arbitrary nonsmooth functions; they are used under the neighborhood or within-domain differentiability assumptions stated in the relevant sections. The bounded-frequency, sparse-node, envelope-size, integer-scale, and optional endpoint conventions remain part of the paper-level model specification whenever the full constrained family is invoked.

The framework is also not a time-stepping scheme, stochastic differential equation solver, Monte Carlo pricing method, or optimal-control algorithm. Those methods may provide downstream financial applications or validation settings, but the present contribution is the local derivative-symbol layer used to construct and evaluate derivative-sensitive features.

These limitations do not affect the internal calculus. They delimit the intended empirical interpretation of the framework.

12 Conclusion

This paper develops a factorized derivative calculus for local oscillatory-envelope turning models. For a node

$$Z(\tau) = b\tau^m e^{\alpha(\tau)} e^{\mathrm{i}(\omega\tau+\phi)},$$

each derivative admits the representation

$$Z^{(n)}(\tau) = Z(\tau)P_n(\tau),$$

where the derivative symbols satisfy $P_{n+1} = P_n' + qP_n$ with $q = m/\tau + \alpha' + \mathrm{i}\omega$. This gives a Bell-polynomial structure in the general case. Under constant envelope slope, the derivative hierarchy collapses to a binomial falling-factorial law.

The resulting derivative blocks provide a closed-form route to local direction, curvature, and curvature-momentum features. These features may support a disciplined local turning-point framework built from sparse nodes, mild envelopes, bounded frequencies, clustered turning-region reporting, and explicit derivative-aware evaluation. For machine computation, the reference implementation avoids early expansion into additive real coordinates and instead uses a quotient package based on log-polar node storage and normalized derivative symbols. The reported synthetic checks are consistent with implementation-level correctness of the quotient-symbol recursion, coordinate equivalences, shifted-coordinate safeguards, and finite-difference validation behavior in double precision. The high-noise stress tests further illustrate the difference between path accuracy and derivative accuracy: level-only fitting can remain visually plausible while higher-derivative errors become large, whereas derivative-aware fitting remains stable in this experiment only in the controlled oracle setting where smooth derivative targets are known. The imaginary-axis coordinate formulation gives an equivalent contour-coordinate package for the oscillatory part, with branch and scaling conventions handled explicitly and without an implicit holomorphic-extension claim. The appropriate role of the model is local sensing and derivative feature construction, not unconstrained global prediction or a direct application-specific forecasting claim.

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